ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

Department of Mathematics, Mathematics for Natural Science

Chapter Three: Functions
Worksheet-III

Course Title: Mathematics for Natural Science
Course Code: Math1007
Academic Year: 2024/25
Semester: First
Year of Study: Freshman
Category: Engineering & Applied

---

1. The Cartesian product $A \times A$ has 9 elements among which $(-1, 0)$ and $(0, 1)$ are found. Find the set A and the remaining elements of $A \times A$.

2. If the ordered pairs $(x^2 - 3x, y^2 + 4y)$ and $(-2, 5)$ are equal, then find x and y.

3. Let R be a relation on the set $A = \{1, 2, 3, 4, 5, 6\}$ defined by $R = \{(a, b) : a + b \leq 11\}$. Then
   (a) list the elements of R
   (b) find R^{-1}
   (c) find Dom(R) and Range(R)
   (d) find Dom(R^{-1}) and Range(R^{-1})

4. Write the domain and range of the following relations. Determine which relations are functions.
   (a) $R_1 = \{(4, 3), (6, 8), (4, 8), (0, 9), (0, 10)\}$
   (b) $R_2 = \left\{\left(\frac{1}{2}, \frac{1}{6}\right), (-1, 1), \left(\frac{1}{3}, \frac{1}{8}\right)\right\}$
   (c) $R_3 = \{(3, 7), (3, 11), (4, 9), (5, 11)\}$
   (d) $R_4 = \{(a, 2), (b, 3), (c, 2), (a, 3), (d, 4), (b, 4)\}$
   (e) $R_5 = \{(-\sqrt{3}, 3), (-1, 1), (0, 0), (1, 1), (\sqrt{3}, 3)\}$
   (f) $R_6 = \{(5, 0), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5)\}$

5. Find the domain and range of the following relations
   (a) $\{(x, y) : x^2 + y^2 \leq 4\}$
   (b) $\{(x, y) : x + y < 1\}$
   (c) $\begin{cases} 2x + y = 1 \\ x + 2y = 1 \end{cases}$
   (d) $\{(x, y) : |x| + y \geq 2\}$

6. Given that
$$f(x) = \begin{cases} \sqrt{x-1}, & x \geq 1 \\ 4, & x < 1. \end{cases}$$
Find
   (a) $f(0)$
   (b) $f(3)$
   (c) $f(a+1)$ in terms of a. (Given that $a \geq 0$)

7. Let $A = \{9, 10, 11, 12, 13, 14, 15, 16, 17\}$, and $f : A \to \mathbb{N}$ be defined by $f(n) =$ the highest prime factor of $n \in A$. Write f as a set of ordered pairs and find the range of f.

8. Find the domain and range of the following functions.
   (a) $f(x) = \sqrt{x^2 - 6x + 8}$
   (b) $f(x) = x^2 + 1, x \in \mathbb{W}$
   (c) $f(x) = \begin{cases} 3x + 4, & -1 \leq x < 2 \\ 1 + x, & 2 \leq x < 5 \end{cases}$
   (d) $f(x) = 2 - x, x \in \mathbb{N}$

9. For $y = (x-1)(x-2)$, state the range if
   (a) the domain is considered to be $\{x \mid 1 \leq x \leq 2\}$
   (b) the domain is considered to be $\{x \mid (-\infty, 1] \cup [2, \infty)\}$

10. For the following functions form fog, gof, and state the domain.
   (a) $f(x) = 2 - \frac{1}{x} + |x|, \quad g(x) = x^2 - 4$
   (b) $f(x) = \sqrt{x}, \quad g(x) = |x|$

11. Let $f(x) = 5x - 3$. Find $g(x)$ so that $(fog)(x) = 2x + 7$.

12. If f is a real function defined by $f(x) = \frac{x-1}{x+1}$, show that
$$f(2x) = \frac{3f(x) + 1}{f(x) + 3}.$$

13. Which of the following functions are one-to-one? onto? and one-to-one correspondence? Explain.
   (a) $f : \mathbb{R} \to \mathbb{R}, f(x) = x^3 - 2x^2 + 1$
   (b) $p : \mathbb{R} \to \mathbb{R}, p(x) = |1 - 3x|$
   (c) $g : [2, \infty) \to \mathbb{R}, g(x) = x^3 - 2x^2 + 1$
   (d) $f : \mathbb{Z} \to \mathbb{Z}, f(n) = n^3 + 1$
   (e) $g : \mathbb{R} \to \mathbb{R}, g(x) = 5^x$
   (f) $f : \mathbb{R} \to \mathbb{R}, f(x) = \sqrt{x^2}$

14. Find the inverse function of $f(x)$. State the domain of the inverse function $f^{-1}(x)$.

    (a) $f(x) = -10 + 8x$
    (b) $f(x) = (x-2)^5 + 3$
    (c) $f(x) = \frac{-2x-2}{x+2}$
    (d) $f(x) = \frac{4-\sqrt[3]{4x}}{2}$
    (e) $f(x) = \sqrt[3]{x+1} + 2$
    (f) $f(x) = \frac{\sqrt[5]{2-x}}{2}$
    (g) $f(x) = \frac{5x+3}{1-2x}$
    (h) $f(x) = -1 + x^3$

15. State which of the following define polynomial functions:

    (a) $y = x^2 + 2^x$
    (b) $y = \frac{4}{x}$
    (c) $y = \sqrt{x}$
    (d) $y = \frac{x+i}{2-i}$
    (e) $f(x) = \sqrt[3]{x+1} + 2$
    (f) $y = x - \sqrt{6x^2 + \pi x^0}$
    (g) $y = (x+2)^{100}$
    (h) $y = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$

16. State which of the following define rational functions:

    (a) $y = \frac{1}{x}$
    (b) $y = |x|$
    (c) $y = \sqrt{x}$
    (d) $y = 1$
    (e) $y = \frac{1-3x^2+x}{x^4-2x^5+1}$
    (f) $y = (-1)^x$
    (g) $y = x + x^2$
    (h) $y = \frac{x-1}{x+1}$
    (i) $y = \frac{\sqrt{2}x+1}{x-1}$

17. Find a polynomial equation of lowest degree which has the given roots:

    (a) 1, 1, 1
    (b) 1, 2, $1+i$, $1-i$
    (c) 1, $-\frac{1}{2} - \frac{i\sqrt{3}}{2}$, $-\frac{1}{2} + \frac{i\sqrt{3}}{2}$

18. Use the Remainder Theorem to find:

    (a) $P(1)$ when $P(x) = x^2 - x + 3$

    (b) $P(i)$ when $P(x) = x^2 + 1$

    (c) $P(0)$ when $P(x) = 3x^3 + x^2 - 5$

19. Given that 3 is a double zero of $P(x) = x^4 - 3x^3 - 19x^2 + 87x - 90$, find all the zeros of $P(x)$.

20. Write the general polynomial $P(x)$ whose only zeros are 1, 2 and 3, with multiplicity 3, 2 and 1 respectively. What is its degree?

21. In the following equations find the rational roots, and, where possible, solve completely.

    (a) $P(x) = x^3 - 2x^2 - 2x + 3 = 0$

    (b) $P(x) = 2x^3 - 3x^2 - 3x + 2 = 0$

    (c) $P(x) = 3x^4 + 2x^3 + 2x^2 - x = 0$

22. How many zeros does the polynomial $x(x - r_1)(x - r_2)(x - r_3) \cdots (x - r_k)$ have?

23. Find the domain and the real zeros of the given functions and sketch their graphs.

    (a) $f(x) = \frac{3x}{x^2+2x-3}$

    (b) $f(x) = \frac{x^3+2x^2-x-2}{x^2+6x+8}$

    (c) $f(x) = \frac{(x+2)(x-3)}{(x+1)^2(x-2)}$

    (d) $f(x) = \frac{(x+2)^2(x-2)}{2(x-1)^2(x-3)}$

24. For what values of k is $x^2 + 2x + k$ exactly divisible by $x - k$?

25. If the polynomial $P(x) = x^5 + Ax^3 + Ax + 4$ is such that $P(2) = 6$, find $P(-2)$.

26. Sketch the graph of the given function. Identify the domain, range, intercepts, and asymptotes.

    (a) $f(x) = \log_2(x - 3)$

    (b) $f(x) = -3 + \log_2 x$

    (c) $f(x) = -\log_x(-3)$

    (d) $f(x) = 3\log_2(x + 5)$

    (e) $f(x) = \log_{\frac{1}{3}} x$

    (f) $f(x) = 2^{-x}$

    (g) $f(x) = 16 - 4^x$

    (h) $f(x) = 1 - e^x$

    (i) $f(x) = e^{x-3}$

27. Solve the following equations.

    (a) $16^{3a-2} = \frac{1}{4}$

    (b) $\log_2(1+\sqrt{x}) = 6$

    (c) $2^x = 8$

    (d) $\log_2(\log_3(x)) = 1$

    (e) $\frac{1}{2}(10^{x-1})^x + 3 = 53$

    (f) $e^{2x} - 7e^x + 10 = 0$

28. Let $f(x) = 3^x$. Show that
$$\frac{f(x+2) - f(2)}{2} = 4(3^x).$$

29. Find the inverse of

    (a) $f(x) = e^{3x-1}$

    (b) $g(x) = \ln\left(\frac{x}{x-1} + 2\right)$, for $x > 1$

30. Sketch the graph of the following functions

    (a) $f(\theta) = 2 + \cos\theta$

    (b) $f(x) = \sin\left(x + \frac{\pi}{3}\right)$

    (c) $f(x) = \tanh(x)$

31. Show that
$$\cos\left(\frac{3}{12}\pi\right) = -\frac{1}{4}\left(\sqrt{2} + \sqrt{6}\right).$$

32. Given $\sin\alpha = -\frac{4}{5}$ (α in fourth quadrant) and $\tan\beta = -\frac{5}{12}$ (β in second quadrant), find $\sin(\alpha - \beta)$ and $\cos(\alpha - \beta)$.

33. Let $\sin 4\theta = \frac{1}{2}$ and $\tan\theta > 0$. Then find

    (a) $\sin 2\theta$

    (b) $\cos 8\theta$

34. Prove that

    (a) $\sin\left(\frac{\pi}{4} - x\right) = \frac{\cos x - \sin x}{\sqrt{2}}$

    (b) $\sin\left(\frac{\pi}{4} - x\right) = \frac{\cos x + \sin x}{\sqrt{2}}$

35. Prove the following identities

    (a) $\sin x(\cot x + \csc x) = \cos x + 1$

    (b) $\sec 2x = \frac{\csc^2 x}{\csc^2 x - 2}$

    (c) $\sinh(x-y) = \sinh(x)\cosh(y) - \cosh(x)\sinh(y)$

    (d) $\sec^2 x - \csc^2 x = \tan^2 x - \cot^2 x$